Table 1: **Reasoning post-training: every checkpoint of Table ??.** Before and After are the checkpoints before and after each OT3-30K update, not before and after Tulu3. Starting checkpoints are evaluated again here, so these values are paired within the table and must not be compared with the Phase 1 tables. Protocol B (Appendix Table 5). Values are mean$_{\pm SD}$ over three sampled runs, and Avg is the mean over the eleven tasks shown.

**(a) Mathematics**

| Model | Stage | A24 | A25 | AMC | M500 | GSM8K |
|---|---|---|---|---|---|---|
| *Main comparison: Tulu3 SFT → OT3-30K* | | | | | | |
| MV | Before | $1.1_{\pm 1.92}$ | $0.0_{\pm 0.00}$ | $8.3_{\pm 3.82}$ | $29.2_{\pm 6.41}$ | $46.3_{\pm 0.88}$ |
| | After | $11.1_{\pm 3.85}$ | $16.7_{\pm 3.33}$ | $46.7_{\pm 6.29}$ | $71.0_{\pm 1.06}$ | $49.0_{\pm 1.75}$ |
| FineWeb-Edu | Before | $0.0_{\pm 0.00}$ | $1.1_{\pm 1.92}$ | $0.0_{\pm 0.00}$ | $0.8_{\pm 0.53}$ | $7.3_{\pm 0.76}$ |
| | After | $0.0_{\pm 0.00}$ | $0.0_{\pm 0.00}$ | $1.7_{\pm 1.44}$ | $0.6_{\pm 0.00}$ | $6.2_{\pm 0.72}$ |
| Nemotron-CC-HQ | Before | $0.0_{\pm 0.00}$ | $0.0_{\pm 0.00}$ | $5.0_{\pm 5.00}$ | $1.9_{\pm 0.31}$ | $5.5_{\pm 0.40}$ |
| | After | $0.0_{\pm 0.00}$ | $0.0_{\pm 0.00}$ | $0.0_{\pm 0.00}$ | $0.3_{\pm 0.12}$ | $2.9_{\pm 0.09}$ |
| Comma0.1 | Before | $0.0_{\pm 0.00}$ | $1.1_{\pm 1.92}$ | $2.5_{\pm 2.50}$ | $1.7_{\pm 0.61}$ | $11.0_{\pm 0.77}$ |
| | After | $0.0_{\pm 0.00}$ | $0.0_{\pm 0.00}$ | $0.0_{\pm 0.00}$ | $1.3_{\pm 0.58}$ | $13.4_{\pm 0.80}$ |
| SmolLM2 | Before | $0.0_{\pm 0.00}$ | $0.0_{\pm 0.00}$ | $1.7_{\pm 2.89}$ | $6.7_{\pm 0.70}$ | $25.9_{\pm 1.45}$ |
| | After | $0.0_{\pm 0.00}$ | $0.0_{\pm 0.00}$ | $1.7_{\pm 1.44}$ | $6.6_{\pm 0.92}$ | $23.1_{\pm 1.34}$ |
| Qwen2.5 | Before | $0.0_{\pm 0.00}$ | $0.0_{\pm 0.00}$ | $6.7_{\pm 1.44}$ | $21.0_{\pm 2.78}$ | $55.1_{\pm 0.39}$ |
| | After | $4.4_{\pm 5.09}$ | $1.1_{\pm 1.92}$ | $26.7_{\pm 7.22}$ | $44.9_{\pm 2.00}$ | $57.6_{\pm 0.34}$ |
| Qwen3 | Before | $1.1_{\pm 1.92}$ | $0.0_{\pm 0.00}$ | $12.5_{\pm 2.50}$ | $24.7_{\pm 2.50}$ | $64.7_{\pm 0.80}$ |
| | After | $0.0_{\pm 0.00}$ | $0.0_{\pm 0.00}$ | $1.7_{\pm 1.44}$ | $10.9_{\pm 0.42}$ | $25.7_{\pm 0.38}$ |
| *MV starting-point controls* | | | | | | |
| 16K extended | Before | $3.3_{\pm 3.33}$ | $8.9_{\pm 3.85}$ | $25.8_{\pm 12.58}$ | $46.5_{\pm 11.29}$ | $55.0_{\pm 1.78}$ |
| | After | $8.9_{\pm 5.09}$ | $25.6_{\pm 1.92}$ | $42.5_{\pm 6.61}$ | $69.8_{\pm 1.25}$ | $58.8_{\pm 1.52}$ |
| Tulu3 SFT + DPO | Before | $1.1_{\pm 1.92}$ | $1.1_{\pm 1.92}$ | $14.2_{\pm 6.29}$ | $30.1_{\pm 3.31}$ | $47.4_{\pm 1.20}$ |
| | After | $0.0_{\pm 0.00}$ | $4.4_{\pm 1.92}$ | $17.5_{\pm 4.33}$ | $34.4_{\pm 1.78}$ | $44.4_{\pm 1.52}$ |

**(b) Code, science and instruction following, and the average**

| Model | Stage | HE | LCB | MBPP | GPQA | JEE | IFE | Avg |
|---|---|---|---|---|---|---|---|---|
| *Main comparison: Tulu3 SFT → OT3-30K* | | | | | | | | |
| MV | Before | $44.9_{\pm 1.76}$ | $7.0_{\pm 2.26}$ | $31.9_{\pm 0.46}$ | $27.4_{\pm 0.58}$ | $7.8_{\pm 1.33}$ | $54.0_{\pm 0.15}$ | $23.4_{\pm 0.31}$ |
| | After | $51.0_{\pm 4.97}$ | $11.5_{\pm 0.87}$ | $28.5_{\pm 3.29}$ | $28.8_{\pm 1.75}$ | $18.8_{\pm 1.34}$ | $42.1_{\pm 0.20}$ | $34.1_{\pm 0.93}$ |
| FineWeb-Edu | Before | $2.2_{\pm 1.96}$ | $0.1_{\pm 0.16}$ | $1.0_{\pm 0.20}$ | $28.6_{\pm 1.27}$ | $3.1_{\pm 0.59}$ | $38.1_{\pm 0.98}$ | $7.5_{\pm 0.37}$ |
| | After | $0.0_{\pm 0.00}$ | $0.0_{\pm 0.00}$ | $0.0_{\pm 0.00}$ | $23.9_{\pm 0.77}$ | $1.9_{\pm 0.44}$ | $33.4_{\pm 0.67}$ | $6.2_{\pm 0.08}$ |
| Nemotron-CC-HQ | Before | $3.5_{\pm 0.93}$ | $0.0_{\pm 0.00}$ | $1.9_{\pm 0.23}$ | $28.5_{\pm 1.77}$ | $3.2_{\pm 0.39}$ | $39.2_{\pm 1.78}$ | $8.1_{\pm 0.89}$ |
| | After | $0.2_{\pm 0.35}$ | $0.0_{\pm 0.00}$ | $0.0_{\pm 0.00}$ | $25.1_{\pm 2.96}$ | $1.1_{\pm 0.20}$ | $35.5_{\pm 0.34}$ | $5.9_{\pm 0.31}$ |
| Comma0.1 | Before | $14.2_{\pm 0.93}$ | $0.4_{\pm 0.31}$ | $16.5_{\pm 2.08}$ | $26.4_{\pm 5.40}$ | $5.5_{\pm 0.72}$ | $46.8_{\pm 0.52}$ | $11.5_{\pm 0.82}$ |
| | After | $1.4_{\pm 0.93}$ | $0.0_{\pm 0.00}$ | $0.1_{\pm 0.12}$ | $26.8_{\pm 2.81}$ | $2.6_{\pm 0.32}$ | $36.8_{\pm 0.08}$ | $7.5_{\pm 0.29}$ |
| SmolLM2 | Before | $23.6_{\pm 2.46}$ | $0.9_{\pm 0.31}$ | $23.3_{\pm 0.46}$ | $25.1_{\pm 0.77}$ | $8.1_{\pm 1.92}$ | $53.8_{\pm 0.72}$ | $15.4_{\pm 0.53}$ |
| | After | $14.2_{\pm 3.01}$ | $0.2_{\pm 0.31}$ | $10.5_{\pm 0.92}$ | $26.3_{\pm 1.01}$ | $3.6_{\pm 0.63}$ | $41.7_{\pm 0.72}$ | $11.6_{\pm 0.16}$ |
| Qwen2.5 | Before | $46.5_{\pm 0.70}$ | $3.1_{\pm 0.31}$ | $39.9_{\pm 1.01}$ | $25.9_{\pm 3.58}$ | $6.9_{\pm 2.23}$ | $54.9_{\pm 0.47}$ | $23.6_{\pm 0.32}$ |
| | After | $21.7_{\pm 6.23}$ | $2.8_{\pm 0.68}$ | $12.3_{\pm 1.79}$ | $25.9_{\pm 0.77}$ | $14.1_{\pm 0.82}$ | $40.6_{\pm 0.98}$ | $22.9_{\pm 0.70}$ |
| Qwen3 | Before | $59.8_{\pm 1.22}$ | $3.4_{\pm 0.41}$ | $46.5_{\pm 1.86}$ | $28.5_{\pm 2.39}$ | $12.7_{\pm 1.00}$ | $59.1_{\pm 1.23}$ | $28.5_{\pm 0.95}$ |
| | After | $8.5_{\pm 1.06}$ | $0.5_{\pm 0.27}$ | $6.7_{\pm 0.76}$ | $29.0_{\pm 2.04}$ | $2.0_{\pm 0.47}$ | $38.8_{\pm 0.98}$ | $11.3_{\pm 0.31}$ |
| *MV starting-point controls* | | | | | | | | |
| 16K extended | Before | $43.7_{\pm 0.93}$ | $8.9_{\pm 0.27}$ | $31.3_{\pm 1.47}$ | $26.9_{\pm 0.29}$ | $10.7_{\pm 1.58}$ | $46.6_{\pm 0.30}$ | $28.0_{\pm 2.73}$ |
| | After | $54.7_{\pm 3.36}$ | $13.4_{\pm 0.87}$ | $28.8_{\pm 1.91}$ | $30.1_{\pm 4.98}$ | $18.4_{\pm 0.41}$ | $40.9_{\pm 1.14}$ | $35.6_{\pm 1.15}$ |
| Tulu3 SFT + DPO | Before | $47.2_{\pm 4.74}$ | $6.7_{\pm 2.67}$ | $34.6_{\pm 1.06}$ | $27.3_{\pm 0.87}$ | $8.2_{\pm 0.75}$ | $54.6_{\pm 1.38}$ | $24.8_{\pm 1.13}$ |
| | After | $46.1_{\pm 1.41}$ | $8.6_{\pm 1.49}$ | $34.5_{\pm 2.44}$ | $27.1_{\pm 1.27}$ | $11.5_{\pm 0.51}$ | $49.2_{\pm 0.61}$ | $25.3_{\pm 0.80}$ |

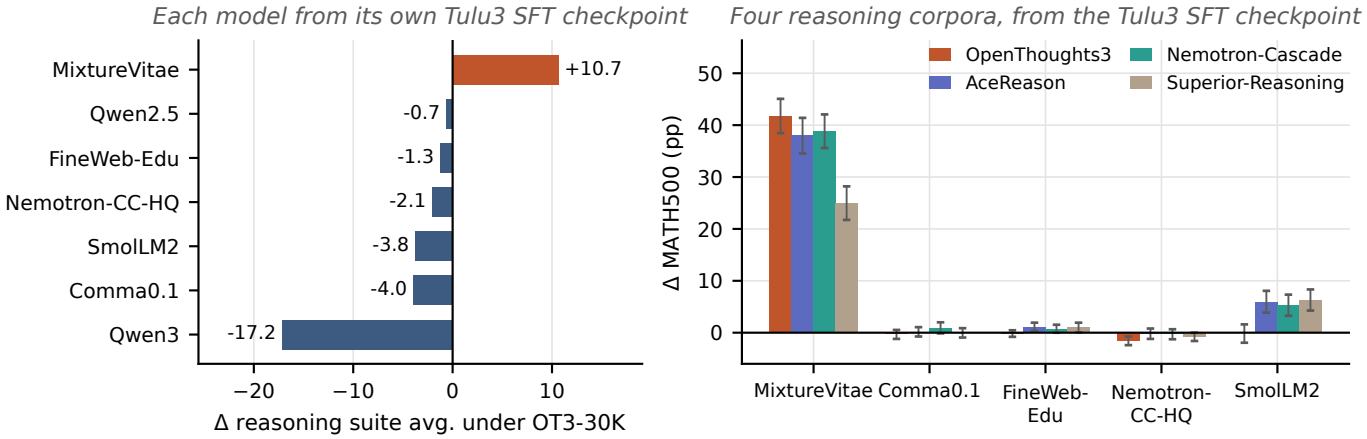
Table 2: **Effect of excluding the OpenThoughts3 (OT3) source from pretraining.** Two 1.7B models pretrained for 300B tokens, with and without the OT3 source, receive the same OT3-30K update directly from their 16K extended checkpoints, without Tulu3. Cells show the mean over three sampled evaluations, with SD below. Avg is the unweighted mean of all eleven benchmarks. Evaluation follows Protocol B.

| | Mathematics | | | | | Code | | | Science/QA | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Stage | **A24** | **A25** | **AMC** | **M500** | **GSM8K** | **HE** | **LCB** | **MBPP** | **GPQA** | **JEE** | **IFE** | **Avg** |
| *With OT3 source* | | | | | | | | | | | | |
| Before | 3.3 ±3.33 | 8.9 ±3.85 | 25.8 ±12.58 | 46.5 ±11.29 | 55.0 ±1.78 | 43.7 ±0.93 | 8.9 ±0.27 | 31.3 ±1.47 | 26.9 ±0.29 | 10.7 ±1.58 | 46.6 ±0.30 | 28.0 ±2.73 |
| After | 8.9 ±5.09 | 25.6 ±1.92 | 42.5 ±6.61 | 69.8 ±1.25 | 58.8 ±1.52 | 54.7 ±3.36 | 13.4 ±0.87 | 28.8 ±1.91 | 30.1 ±4.98 | 18.4 ±0.41 | 40.9 ±1.14 | 35.6 ±1.15 |
| *Without OT3 source* | | | | | | | | | | | | |
| Before | 1.1 ±1.92 | 0.0 ±0.00 | 12.5 ±4.33 | 29.8 ±4.33 | 55.7 ±0.57 | 36.4 ±1.86 | 6.6 ±2.60 | 32.3 ±1.40 | 26.1 ±0.77 | 11.0 ±1.89 | 49.4 ±0.46 | 23.7 ±0.91 |
| After | 2.2 ±1.92 | 3.3 ±0.00 | 14.2 ±3.82 | 38.5 ±1.10 | 58.5 ±0.22 | 39.8 ±3.92 | 5.0 ±1.74 | 33.5 ±1.33 | 27.9 ±1.05 | 10.4 ±1.42 | 46.1 ±0.46 | 25.4 ±0.50 |

**Suite-average gain:** with the OT3 source, +7.6, without it, +1.7 points.

Table 3: **MATH500 changes on all 500 problems.** Change in MATH500 accuracy when each base's Tulu3 SFT checkpoint receives 30K samples from one of four reasoning corpora, on every problem, including those identified in the training data. Table **??** gives the same changes without them. Each problem is scored as its mean over three sampled runs. Brackets are 95% paired bootstrap intervals over the same 500 problems. **Bold** marks a significant increase and *italic* a significant decrease after Benjamini–Hochberg correction over the 20 changes. Protocol B (Appendix Table 5).

| **Base** | **OpenThoughts3** | **AceReason** | **Nemotron-Cascade** | **Superior-Reasoning** |
|---|---|---|---|---|
| MixtureVitae | **+41.8** [+38.5, +45.1] | **+38.0** [+34.5, +41.4] | **+38.9** [+35.6, +42.1] | **+24.9** [+21.7, +28.2] |
| Comma0.1 | −0.3 [−1.2, +0.5] | +0.1 [−0.7, +1.1] | +0.9 [−0.1, +2.0] | +0.0 [−0.9, +0.9] |
| FineWeb-Edu | −0.2 [−0.8, +0.5] | **+1.1** [+0.3, +1.9] | +0.7 [+0.0, +1.5] | +1.0 [+0.1, +1.9] |
| Nemotron-CC-HQ | *−1.5* [−2.4, −0.7] | −0.2 [−1.2, +0.8] | −0.3 [−1.3, +0.7] | −0.8 [−1.6, +0.0] |
| SmolLM2 | −0.1 [−1.9, +1.6] | **+5.9** [+3.9, +8.1] | **+5.3** [+3.3, +7.3] | **+6.3** [+4.3, +8.3] |




phase2_receptivity.pdf

Table 4: Full grid of reasoning corpora for MixtureVitae: four reasoning corpora applied from three starting checkpoints, with recipe, sample budget, and epoch count held fixed so that only the identity of the reasoning corpus changes. The change in parentheses is against the block's starting checkpoint. Protocol B (Appendix Table 5). Values are mean$_{\pm SD}$ over three sampled runs, and Avg is the mean over the eleven tasks shown.

**(a) Mathematics**

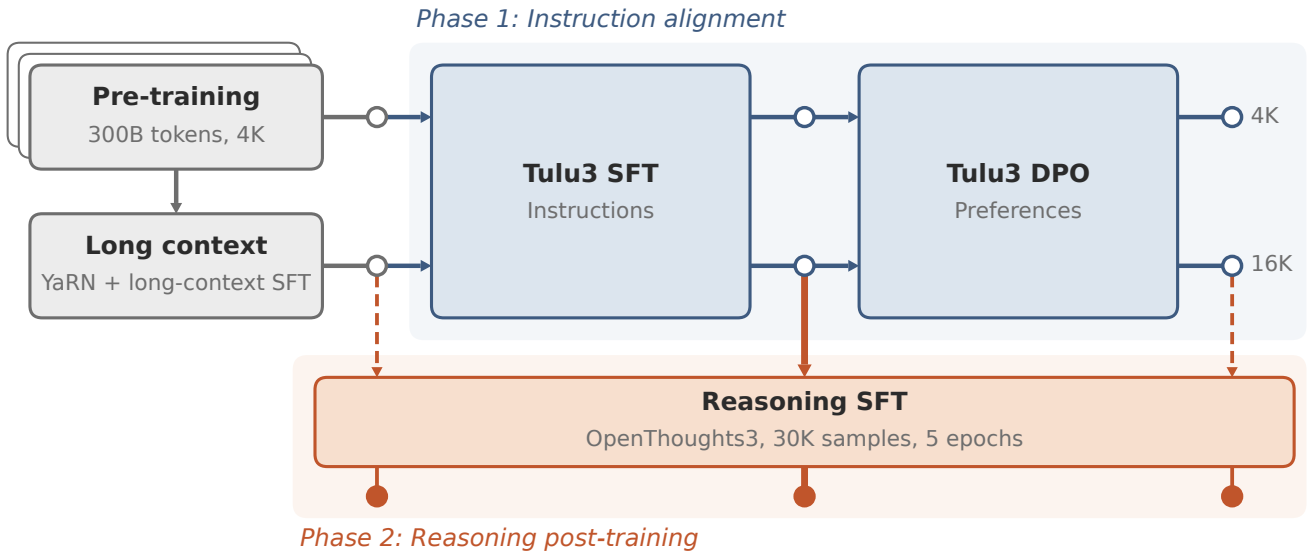
| **Reasoning corpus** | **A24** | **A25** | **AMC** | **M500** | **GSM8K** |
|---|---|---|---|---|---|
| *Starting from the 16K extended checkpoint* | | | | | |
| *starting checkpoint* | $3.3_{\pm 3.33}$ | $8.9_{\pm 3.85}$ | $25.8_{\pm 12.58}$ | $46.5_{\pm 11.29}$ | $55.0_{\pm 1.78}$ |
| OpenThoughts3 | $8.9_{\pm 5.09}$ | $25.6_{\pm 1.92}$ | $42.5_{\pm 6.61}$ | $69.8_{\pm 1.25}$ | $58.8_{\pm 1.52}$ |
| AceReason | $6.7_{\pm 6.67}$ | $13.3_{\pm 0.00}$ | $45.0_{\pm 2.50}$ | $68.0_{\pm 1.71}$ | $59.7_{\pm 0.65}$ |
| Nemotron-Cascade | $11.1_{\pm 8.39}$ | $14.4_{\pm 1.92}$ | $44.2_{\pm 1.44}$ | $67.0_{\pm 1.11}$ | $59.9_{\pm 0.31}$ |
| Superior-Reasoning | $4.4_{\pm 1.92}$ | $6.7_{\pm 0.00}$ | $32.5_{\pm 2.50}$ | $51.4_{\pm 0.53}$ | $63.0_{\pm 0.74}$ |
| *Starting from the Tulu3 SFT checkpoint* | | | | | |
| *starting checkpoint* | $1.1_{\pm 1.92}$ | $0.0_{\pm 0.00}$ | $8.3_{\pm 3.82}$ | $29.2_{\pm 6.41}$ | $46.3_{\pm 0.88}$ |
| OpenThoughts3 | $11.1_{\pm 3.85}$ | $16.7_{\pm 3.33}$ | $46.7_{\pm 6.29}$ | $71.0_{\pm 1.06}$ | $49.0_{\pm 1.75}$ |
| AceReason | $10.0_{\pm 0.00}$ | $15.6_{\pm 1.92}$ | $45.0_{\pm 4.33}$ | $67.2_{\pm 0.60}$ | $57.2_{\pm 0.62}$ |
| Nemotron-Cascade | $11.1_{\pm 1.92}$ | $16.7_{\pm 3.33}$ | $44.2_{\pm 1.44}$ | $68.1_{\pm 0.81}$ | $72.0_{\pm 0.18}$ |
| Superior-Reasoning | $4.4_{\pm 1.92}$ | $7.8_{\pm 5.09}$ | $28.3_{\pm 5.20}$ | $54.1_{\pm 1.30}$ | $57.8_{\pm 0.55}$ |
| *Starting from the Tulu3 SFT + DPO checkpoint* | | | | | |
| *starting checkpoint* | $1.1_{\pm 1.92}$ | $1.1_{\pm 1.92}$ | $14.2_{\pm 6.29}$ | $30.1_{\pm 3.31}$ | $47.4_{\pm 1.20}$ |
| OpenThoughts3 | $0.0_{\pm 0.00}$ | $4.4_{\pm 1.92}$ | $17.5_{\pm 4.33}$ | $34.4_{\pm 1.78}$ | $44.4_{\pm 1.52}$ |
| AceReason | $12.2_{\pm 1.92}$ | $16.7_{\pm 0.00}$ | $40.8_{\pm 6.29}$ | $67.1_{\pm 1.33}$ | $57.3_{\pm 0.26}$ |
| Nemotron-Cascade | $7.8_{\pm 1.92}$ | $18.9_{\pm 1.92}$ | $43.3_{\pm 7.64}$ | $68.7_{\pm 1.51}$ | $71.8_{\pm 0.36}$ |
| Superior-Reasoning | $6.7_{\pm 3.33}$ | $6.7_{\pm 0.00}$ | $29.2_{\pm 3.82}$ | $53.5_{\pm 1.81}$ | $57.6_{\pm 0.38}$ |

**(b) Code, science and instruction following, and the average**

| **Reasoning corpus** | **HE** | **LCB** | **MBPP** | **GPQA** | **JEE** | **IFE** | **Avg** |
|---|---|---|---|---|---|---|---|
| *Starting from the 16K extended checkpoint* | | | | | | | |
| *starting checkpoint* | $43.7_{\pm 0.93}$ | $8.9_{\pm 0.27}$ | $31.3_{\pm 1.47}$ | $26.9_{\pm 0.29}$ | $10.7_{\pm 1.58}$ | $46.6_{\pm 0.30}$ | $28.0_{\pm 2.73}$ |
| OpenThoughts3 | $54.7_{\pm 3.36}$ | $13.4_{\pm 0.87}$ | $28.8_{\pm 1.91}$ | $30.1_{\pm 4.98}$ | $18.4_{\pm 0.41}$ | $40.9_{\pm 1.14}$ | $35.6_{\pm 1.15}$ (+7.6) |
| AceReason | $58.3_{\pm 0.70}$ | $13.8_{\pm 0.27}$ | $37.5_{\pm 1.89}$ | $29.8_{\pm 3.15}$ | $20.6_{\pm 0.88}$ | $40.2_{\pm 1.24}$ | $35.7_{\pm 0.72}$ (+7.7) |
| Nemotron-Cascade | $54.7_{\pm 2.82}$ | $12.6_{\pm 0.78}$ | $37.7_{\pm 1.63}$ | $28.6_{\pm 1.27}$ | $20.6_{\pm 0.60}$ | $43.1_{\pm 0.87}$ | $35.8_{\pm 0.68}$ (+7.8) |
| Superior-Reasoning | $31.5_{\pm 5.67}$ | $8.6_{\pm 0.41}$ | $20.6_{\pm 1.06}$ | $25.9_{\pm 1.62}$ | $10.4_{\pm 1.03}$ | $41.6_{\pm 0.61}$ | $27.0_{\pm 0.34}$ (−1.0) |
| *Starting from the Tulu3 SFT checkpoint* | | | | | | | |
| *starting checkpoint* | $44.9_{\pm 1.76}$ | $7.0_{\pm 2.26}$ | $31.9_{\pm 0.46}$ | $27.4_{\pm 0.58}$ | $7.8_{\pm 1.33}$ | $54.0_{\pm 0.15}$ | $23.4_{\pm 0.31}$ |
| OpenThoughts3 | $51.0_{\pm 4.97}$ | $11.5_{\pm 0.87}$ | $28.5_{\pm 3.29}$ | $28.8_{\pm 1.75}$ | $18.8_{\pm 1.34}$ | $42.1_{\pm 0.20}$ | $34.1_{\pm 0.93}$ (+10.7) |
| AceReason | $54.7_{\pm 1.41}$ | $14.2_{\pm 1.10}$ | $36.1_{\pm 0.99}$ | $26.1_{\pm 1.17}$ | $20.9_{\pm 1.05}$ | $41.6_{\pm 0.34}$ | $35.3_{\pm 0.47}$ (+11.9) |
| Nemotron-Cascade | $55.1_{\pm 0.93}$ | $13.6_{\pm 1.76}$ | $37.2_{\pm 1.59}$ | $28.1_{\pm 0.58}$ | $20.9_{\pm 1.14}$ | $44.1_{\pm 0.98}$ | $37.4_{\pm 0.34}$ (+13.9) |
| Superior-Reasoning | $29.5_{\pm 1.86}$ | $7.8_{\pm 0.56}$ | $21.8_{\pm 0.35}$ | $25.3_{\pm 2.31}$ | $10.2_{\pm 1.06}$ | $41.2_{\pm 0.61}$ | $26.2_{\pm 0.48}$ (+2.7) |
| *Starting from the Tulu3 SFT + DPO checkpoint* | | | | | | | |
| *starting checkpoint* | $47.2_{\pm 4.74}$ | $6.7_{\pm 2.67}$ | $34.6_{\pm 1.06}$ | $27.3_{\pm 0.87}$ | $8.2_{\pm 0.75}$ | $54.6_{\pm 1.38}$ | $24.8_{\pm 1.13}$ |
| OpenThoughts3 | $46.1_{\pm 1.41}$ | $8.6_{\pm 1.49}$ | $34.5_{\pm 2.44}$ | $27.1_{\pm 1.27}$ | $11.5_{\pm 0.51}$ | $49.2_{\pm 0.61}$ | $25.3_{\pm 0.80}$ (+0.5) |
| AceReason | $52.8_{\pm 7.74}$ | $14.8_{\pm 0.87}$ | $36.7_{\pm 0.50}$ | $29.0_{\pm 2.59}$ | $20.7_{\pm 0.86}$ | $42.1_{\pm 1.66}$ | $35.5_{\pm 0.71}$ (+10.7) |
| Nemotron-Cascade | $52.8_{\pm 4.06}$ | $14.4_{\pm 0.72}$ | $37.7_{\pm 1.62}$ | $28.6_{\pm 0.77}$ | $22.2_{\pm 0.44}$ | $43.9_{\pm 1.43}$ | $37.3_{\pm 1.03}$ (+12.5) |
| Superior-Reasoning | $29.7_{\pm 1.41}$ | $6.2_{\pm 0.00}$ | $22.9_{\pm 1.33}$ | $24.6_{\pm 2.28}$ | $10.1_{\pm 1.77}$ | $41.2_{\pm 0.15}$ | $26.2_{\pm 1.18}$ (+1.4) |

Table 5: Protocol B, the reasoning post-training protocol. Used for every OpenThoughts3-30K experiment, namely the endpoints of Table **??**, the pretraining ablation of Table 2, and the grid of reasoning corpora of Tables **??** and 4. Every task is sampled at temperature 0.7, with one answer per question in each of three runs, and results are the mean and SD over the runs.

| Abbr. | Task | Metric | Questions | Max new toks |
|---|---|---|---|---|
| A24 | AIME24 | Accuracy | 30 | 32,000 |
| A25 | AIME25 | Accuracy | 30 | 32,000 |
| AMC | AMC23 | Accuracy | 40 | 32,000 |
| M500 | MATH500 | Accuracy | 500 | 32,000 |
| GSM8K | GSM8K | Exact match, flexible answer extraction | 1,319 | 16,000 |
| HE | HumanEval | Pass@1, Python problems | 164 | 16,000 |
| LCB | LiveCodeBench | Accuracy | 369 | 16,000 |
| MBPP | MBPP | Pass@1 | 500 | 16,000 |
| GPQA | GPQA-Diamond | Accuracy | 198 | 32,000 |
| JEE | JEEBench | Accuracy | 515 | 32,000 |
| IFE | IFEval | Instruction-level accuracy, strict | 500 | 16,000 |




figure_pipeline.pdf